METRIC

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6 August 1993
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26 October 1987
(See 6.7)

MILITARY SPECIFICATION

BREATHING APPARATUS, OXYGEN GENERATING, TYPE A-4 (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one classification of a self-generating, type A-4 oxygen breathing apparatus for use by firefighting, damage control, and other personnel.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-17671 - Canister, Oxygen Breathing Apparatus, for Type A-4.
MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated.

STANDARDS

FEDERAL
FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-151 - Metals: Test Methods.
FED-STD-191 - Textile Test Methods.
FED-STD-595 - Colors Used In Government Procurement.
FED-STD-601 - Rubber: Sampling and Testing.

MILITARY
MIL-STD-2073-1 - DOD Material Procedures for Developing and Application of Packaging Requirements.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 03Q42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
803-5184270 - Oxygen Breathing Apparatus Navy Type A-4, Assemblies And Details.
803-5184258 - Oxygen Breathing Apparatus, Canister.
804-5773272 - Lockers, Oxygen Apparatus Shelf Unit.

(Application for copies should be addressed to: Commander, Portsmouth Naval Shipyard, Code 202.2, Portsmouth, NH 03801.)

PUBLICATION

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)
NBS Circular Special 374 - Methods of Determining the Resolving Power of Photographic Lenses.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 NonGovernment publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 412 - Standard Test Methods for Rubber Properties in Tension. (DOD adopted)
D 413 - Standard Test Methods for Rubber Property - Adhesion to Flexible Substrate. (DOD adopted)
D 1003 - Standard Test Method for Haze and Luminous Transmittance of Transparent Plastics. (DOD adopted)
D 1149 - Standard Test Method for Rubber Deterioration - Surface Cracking in an Ozone Chamber.
D 3935 - Standard Specification for Polycarbonate (PC) Unfilled and Reinforced Material.
D 3951 - Standard Practice for Commercial Packaging. (DOD adopted)
D 4727 - Standard Specifications for Corrugated and Solid Fiberboard Sheet Stock (Container Grade) and Cut Shapes. (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(NonGovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Materials.

3.2.1 Composition. The material shall be as specified herein and in accordance with Drawing 803-5184270. Metal parts shall be corrosion-resistant or coated to protect them from corrosion; cadmium plating shall not be used. Elastomeric parts shall be resistant to deterioration caused by ocean environment and shall be nonirritating to the user of the apparatus.

3.2.2 Rubber compound. The rubber compound shall be 70 percent natural rubber and 30 percent nitrile, plus or minus 2 percent. The nitrile component shall be a copolymer of butadiene and acrylonitrile. The rubber shall contain no mineral sulphate or lead. It shall contain an antioxidant compound and shall be resistant to ozone attack (see 4.6.1). The rubber compound shall show no irritation to the skin (see 4.6.1.2). The finished rubber shall be black in color in accordance with FED-STD-595 and shall have the characteristics shown in table I (see 4.6.1.1).

TABLE I. Physical characteristics of rubber compound.

Characteristic	Before aging	After aging (percent change)	
Tensile strength (minimum)	1.684 x 10^6 Kg/m^2 2/ 2395 psi	+20	-25
Ultimate elongation (percent, min)	550	+20	-15
Permanent set 1/ (percent, maximum)	20	+20	--

1/ After elongation to 75 percent of the ultimate elongation for 10 minutes and resting for 10 minutes.
2/ Kilograms per square meter (kg/m^2). Pounds per square inch (psi).

3.2.3 Neoprene-coated fabrics. The rubber compound used on the fabric shall be a polymer of chloroprene (2-chloro-1, 3-butadiene). The coated fabric shall be made of cotton and shall be in accordance with Drawing 803-5184270. It shall be black in color in accordance with FED-STD-595 and the neoprene shall have the physical characteristics shown in table II (see 4.6.1.1).

TABLE II. Physical characteristics of neoprene rubber.

Characteristic	Before aging	After aging
Minimum tensile strength	1.14 X 10^6 kg/m^2 1621.5 psi	8.437 X 10^5 kg/m^2 1200 psi
Ultimate elongation (minimum) (percent)	750	635

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3.2.4 Self-extinguishing material. The coated fiberglass cloth in accordance with Drawing 803-5184270 shall flame out within 2 seconds (see 4.6.8). The exposed side of the cloth shall be black in color in accordance with FED-STD-595.

3.2.5 Threads. Threads shall be in accordance with FED-STD-H28 and Drawing 803-5184270.

3.2.6 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 Assembly. The breathing apparatus shall be a closed circuit and self-contained in accordance with Drawing 803-5184270. The apparatus shall ensure the maximum safety and comfort of the wearer.

3.4 Construction and maintenance. Any part of the apparatus that is in direct contact with the wearer's body shall be of nonirritating composition. The apparatus shall contain the following principal parts:

(a) Mask assembly
  (1) facepiece
  (2) facepiece form
  (3) lens
  (4) lens cover/protector[1]
  (5) lens retaining ring
  (6) nosecup assembly
  (7) speaking diaphragm assembly
  (8) speaking diaphragm wrench
  (9) combination valve
  (10) head harness
  (11) buckles
  (12) breathing tubes
(b) Body assembly
  (1) plunger assembly
  (2) canister guard and breastplate
  (3) retaining assembly
  (4) breathing bags
  (5) relief valve assembly
  (6) timer assembly
(c) Harnesses
  (1) waist belt
  (2) harness assembly

[1] For clarification, the lens protector is an adhesive backed paper to protect lens during shipping. The protector must be easily removed. Standard commercial protector is acceptable.

3.5 Performance characteristics. Performance characteristics shall be as specified in 3.5.1 through 3.5.7.

3.5.1 Breathing resistance. Resistance to gas flow of an apparatus complete with a dummy canister shall not exceed 75 millimeters (mm) (2.95 inches) of water column height (see 4.6.3).

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NOTE: The term "dummy canister" in this specification refers to a canister of the same weight and dimensions as a canister made in accordance with MIL-C-17671, but which is not operational. This maybe a spent canister or a canister which is specially constructed for testing purposes.

3.5.2 Leakage. There shall be no leakage in excess of the amounts specified in 3.6.1 for the facepiece assembly, 3.6.2 for the body, and 3.6.2.2.1 for the relief valve assembly (see 4.6.9 through 4.6.9.3).

3.5.3 Corrosion resistance. Immediately after exposure to the salt spray atmosphere specified in 4.6.4, the apparatus shall have no visible sign of corrosion on the operating mechanism, canister guide, or facepiece. The apparatus shall exhibit no changes which could prevent the equipment from meeting the performance requirements of this specification.

3.5.4 High impact shock. The apparatus shall meet the leakage requirements of 3.6.1 and 3.6.2 (see 4.6.5).

3.5.5 Impact. The apparatus body shall be dropped as specified in 4.6.6 and shall be fully operable without a facepiece attached.

3.5.6 Wear. Repeated operation of the bail as specified in 4.6.7 shall not degrade the operation of the apparatus.

3.5.7 Apparatus output. The apparatus, when used in conjunction with a canister which is in accordance with MIL-C-17671, shall supply breathable gas, as defined in MIL-C-17671, for a 50-minute minimum. The temperature of the breathable gas shall not exceed 46 degrees Celsius (°C) (115 degrees Fahrenheit (°F)) (see 4.6.10). The average carbon dioxide volumetric content of the gas received from the apparatus shall not exceed 1.5 percent nor shall the carbon dioxide volumetric content of the gas exceed 5.0 percent at any time (see 4.6.10).

3.6 Components. Leakage requirements shall be as specified in 3.5.2 (see 4.6.9 through 4.6.9.4).

3.6.1 Facepiece assembly. The facepiece assembly shall be in accordance with Drawing 803-5184270. It shall include the facepiece, hoses, and valves necessary to bring breathable gas from the body of the apparatus to the respiratory passages of the wearer and to conduct exhaust gas back to the body of the apparatus. The allowable leakage shall be 0.02 percent by volume, during the 15 seconds after the test has started (see 4.6.9). The components of the facepiece assembly shall be in accordance with Drawing 803-5184270 and the following as specified (see 4.6.9.2).

3.6.1.1 Facepiece. The facepiece shall be made of molded rubber as specified in 3.2.2. The facepiece shall have one lens.

3.6.1.1.1 Facepiece form. A plastic facepiece form in accordance with Drawing 803-5184270 shall be installed in each facepiece to ensure protection of the facial seal during storage.

3.6.1.2 Lens. Each facepiece shall be fitted with a lens molded in accordance with Drawing 803-5184270 from a colorless polycarbonate resin in accordance with ASTM D 3935. The lens shall not crack, fracture, chip, or dislodge from the retaining rings (see 4.6.2.3).

3.6.1.2.1 Optical properties. The finished lens shall be free of striae scratches, discoloration, or other defects interfering with clearness of vision or causing eyestrain under prolonged use. Standards in this respect are those used for ophthalmic lenses and not those used for high precision optical devices. The lens shall transmit not less than 85 percent of light incident on the surface. The lens shall not be positive in refractive power and shall not have a negative refractive power greater than 1/4 diopter (see 4.6.2.4).

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3.6.1.2.2 Lens construction. The outer cylindrical surface of the lens shall be molded to the radii in accordance with Drawing 803-5184270. The inner surface shall be molded to a cylindrical form calculated to accomplish an error refraction within the limits stated in table III. These limits shall apply to all points on the lenses inside a line, parallel to the axis of curvature, and subtending at an angle of 37 degrees with the optical axis (see 4.6.2.2).

TABLE III. Lens construction details.

Cylindrical powers1/	Convexity zero	Concavity 1/4 diopter (Max)
Prism power, vertical	Superconducting 1/7 diopter	Subducting 1/8 diopter
Prism power, horizontal diopters	Abducting zero	Abducting 1-1/2

1/ These limits are applicable to all points on the lens inside a line parallel to the periphery of the lens, which is a distance of 12.7 mm (0.5 inch) from the edge.

3.6.1.3 Lens cover and protector. A self-adhesive lens protector shall be provided to protect the lenses during shipping. The protector shall not degrade lens performance below the requirements of this specification. The protector shall be easily removable.

3.6.1.4 Lens retaining rings. The plastic lens shall be retained in place in the molded opening in the facepiece by means of lens retaining rings. The retaining rings shall be of a molded material and made so that when in place they shall compress the rubber under the solid part of the retaining ring forming a gas-tight seal.

3.6.1.5 Nosecup assembly. A nosecup assembly shall channel the moist, exhaled gas to the exit port in the combination valve. The nosecup assembly shall be installed so that it will not restrict breathing or visual lines of sight.

3.6.1.6 Speaking diaphragm assembly. The facepiece shall be equipped with a speaking diaphragm assembly.

3.6.1.6.1 Diaphragm wrench. A wrench for removing the cover over the speaking diaphragm shall be provided in the shipping container.

3.6.1.7 Combination valve. The facepiece shall be equipped with a combination valve body that directs the flow of exhalation and inhalation gas. The valve shall be of a sturdy leakproof construction, and shall direct breathable gas from the body of the apparatus toward the wearer, and exhaust gas to the body of the apparatus.

3.6.1.8 Head harness. The finished head harness shall be made of molded rubber as specified in 3.2.2 and shall be molded not greater than 12 months before delivery. The head harness shall be adjustable and shall maintain tension for 1 hour. The rubber of the head harness shall be black in color in accordance with FED-STD-595 and shall have physical characteristics as specified (see 3.8 and 4.6.1.1).

3.6.1.8.1 Buckles. The facepiece shall be equipped with buckles in accordance with Drawing 803-5184270. The buckles shall be compatible with the head harness straps, shall be constructed of stainless steel, and shall be the quick-adjustable, roller type.

3.6.1.9 Breathing tubes. The facepiece shall be equipped with two breathing tubes, one for inhalation and one for exhalation. The breathing tubes shall be constructed from rubber as specified in 3.2.2 and shall be gas-tight (see 4.6.11).

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3.6.1.9.1 Breathing tube fittings. The ends of the breathing tubes shall terminate in quick-connect fittings that match those on the body of the device. The inhalation and exhalation hoses shall have different size fittings to prevent the wearer from attaching the hoses to the wrong ports during operation of the device.

3.6.2 Body. The body of the apparatus shall be in accordance with Drawing 803-5184270. The body shall include a canister holder, breathing bags, and a timer. The allowable leakage of the body (see 4.6.9) shall be 3.05 centimeters (cm) (1.2 inches) of water column height within the first minute of measurement. The minimum setting of relief valve (see 4.6.9.1) shall be 1.90 cm (0.75 inches) of water column. The components of the body shall be in accordance with Drawing 803-5184270.

3.6.2.1 Canister holder. The canister holder shall consist of the plunger assembly, canister guard, breastplate, and retaining system. In accordance with Drawing 803-5184258, these parts shall be assembled to permit the canister to be inserted and held in a standby position without puncturing the canister seal. Operation of the retaining assembly shall ram the canister home on the plunger assembly, piercing the canister seal and shall gas-tight seal the canister to the apparatus at the plunger assembly.

3.6.2.1.1 Plunger assembly. The canister holder shall be equipped with a plunger assembly that shall pierce the canister seal in accordance with MIL-C-17671.

3.6.2.1.2 Canister guard and breastplate. The canister holder shall be equipped with a canister guard and breastplate that shall be covered with an insulating, self-extinguishing (see 3.2.4) material on all externally exposed surfaces.

3.6.2.1.3 Retaining system. While in the standby position, the retaining system shall allow easy insertion of the oxygen breathing apparatus canister in accordance with MIL-C-17671. The retaining system shall have a handle that shall latch in the standby position to retain the canister without puncturing the (copper foil) seal (see 4.6.6). Movement of the handle to the latches (operational) position shall lift the canister to a seated position with the plunger housing. The canister spring (in the retaining system) shall exert sufficient force to prohibit leakage between the canister and the plunger housing (see 4.6.9).

3.6.2.2 Breathing bags. The breathing bags shall be constructed of neoprene coated fabric and shall have the physical characteristics as specified in 3.2.3. The bags shall be impermeable to air at pressures up to 25.4 cm (10 inches) of water column height (see 4.6.9.1). The breathing bags as installed in the apparatus body shall have a capacity of 5 liters (0.18 cubic feet (ft^3)) minimum (see 4.6.14). The vulcanized seam joining the front and rear halves of the breathing bag shall be adequate for 250,000 cycles of normal operation (see 4.6.15.1). The seam shall have a minimum peel strength of 12.0 pounds per inch at 121°C (250°F) (see 4.6.15.2).

3.6.2.2.1 Relief valve assembly. The left breathing bag shall be fitted with a relief valve. The valve shall operate by means of an internal mechanical linkage when the bag expands. The valve shall incorporate a check valve to prevent entry of ambient air into the apparatus. The relief valve shall be spring loaded with sufficient tension so that when in the closed position, leakage will not exceed 0.15 milliliter per minute (mL/min) (0.0061 cubic inch per minute (in/min) (see 4.6.9.3).

3.6.2.3 Timer assembly. A timer assembly shall be installed on the body of the apparatus. The timer shall sound a warning signal in the form of a continuous ring of 10 seconds duration (minimum) (see 4.6.12.2). This signal shall operate after the timer has been fully wound and shall be set for any time period up to 1 hour. The timer shall be spring activated, and the spring shall be made of materials resistant to salt water corrosion. One control knob shall wind, set, and activate the timer. The timer shall deviate not greater than plus or minus 5 minutes in 1 hour (see 4.6.12.3). The sound intensity of the ring shall be 75

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decibels A-weighted (dBA) minimum (see 4.6.12.2). Water seepage into timer body shall not exceed 0.2 mL (see 4.6.12.1). The control knob shall withstand a shaft torque of 35 inch-pounds (in-lb) without breaking or deforming (see 4.6.12.4).

3.6.3 Harnesses. Two harnesses shall be supplied with the apparatus, the waist strap assembly, and the harness assembly.

3.6.3.1 Waist strap assembly. The waist strap assembly shall be as shown on Drawing 803-5184270.

3.6.3.2 Harness assembly. The main harness assembly shall be as shown on Drawing 803-5184270.

3.7 Weight of apparatus. The weight of the apparatus shall not exceed 7.49 kilograms (16.5 pounds) without the canister (see 4.6.13).

3.8 Color. Where black is the specified color, it shall be in accordance with FED-STD-595 in a range, color 27038 to 27040 (semi-gloss black) or 37038 to 37056 (lusterless black). Other colors shall be in accordance with Drawing 803-5184270.

3.9 Finish. Finishes shall be in accordance with Drawing 803-5184270.

3.10 Identification. The apparatus shall be identified by a nameplate and a serial number. The nameplate shall be located on the plunger assembly as shown on Drawing 803-5184270. The nameplate shall contain the following information: "Navy Oxygen Breathing Apparatus, Type A-4", the Julian date of manufacture, manufacturer's Commercial Activity Government Entity (CAGE) number, contract number, and National stock number. The serial number shall be attached to the wearer's side of the breastplate, to the wearer's right, as shown on Drawing 803-5184270. The combination valve shall be marked with a decal stating "Install This Side UP", in accordance with Drawing 803-5184270. A date code shall be imprinted on the facepiece, head harness, breathing tubes, nosecup, and breathing bag connector tube in accordance with Drawing 803-5184270.

3.11 Workmanship. The apparatus shall be uniform in quality and shall be free from irregularities, defects, or foreign matter that could affect safety, performance, reliability, or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. Four complete breathing apparatuses and two additional breathing bags shall be provided for first article inspection. The sample apparatuses shall be subjected to the examinations and tests as specified in 4.3.1 using the same methods, materials, and equipment as those which shall be used during regular production. In order to avoid destruction of the apparatus, sample pieces of nitril rubber, neoprene, and coated fiberglass cloth shall be used as specified in 4.3.2. If any sample fails any examination or test, it shall constitute failure of first article inspection. Any sample containing a critical defect (see 4.4.2.4) fails first article test.

4.3.1 First article inspection procedure. First article inspection procedure shall be as specified in 4.3.1.1 and 4.3.1.2.

4.3.1.1 Examination and nondestructive tests. The four breathing apparatuses furnished for first article inspection shall be subjected to the examinations and tests specified in table IV.

TABLE IV. First article inspection (examinations and nondestructive tests).

Examination or test	Requirement	Test Method	Sample quantity
Apparatus weight	3.7	4.6.13	4
Visual and dimensional examination	3.2.5, 3.3, 3.4, 3.6, 3.8, 3.9, 3.10, 3.11	4.5.1, 4.5.2	4
Lens examination	3.6.1.2.1	4.5.3	4
Lens tests			
Visual characteristics test	3.6.1.2.1	4.6.2.1	4
Light transmission test	3.6.1.2.1	4.6.2.4	4
Resistance to gas flow	3.5.1	4.6.3	4
Wear test	3.5.6	4.6.7	4
Leakage tests	3.5.2	4.6.9	4
Apparatus body test	3.6.2	4.6.9.1	4
Facepiece test	3.6.1	4.6.9.2	4
Relief valve test	3.6.2.2.1	4.6.9.3	4
Breathing bag test	3.6.2.2	4.6.9.4	4
Apparatus output	3.5.7	4.6.10	4
Breathing tube test	3.6.1.9	4.6.11	4
Timer loudness and duration test	3.6.2.3	4.6.12.2	4
Breathing bag volume test	3.6.2.2	4.6.14	4
Timer accuracy	3.6.2.3	4.6.12.3	4
Timer knob strength	3.6.2.3	4.6.12.4	4

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4.3.1.2 For destructive tests. The four breathing apparatuses furnished for first article inspection shall be inspected in accordance with 4.3.1.1 and then be subjected to the tests specified in table V. The two additional breathing bags furnished for first article inspection shall be used for the seam tensile peel test.

TABLE V. First article inspection (destructive tests).

Test	Requirement	Test method	Sample quantity
Lens tests			
Cylinder power test	3.6.1.2.2	4.6.2.2	4
Strength and impact resistance test	3.6.1.2	4.6.2.3	
Rubber compound test			
Oven aging	3.2.2, 3.2.3	4.6.1.1	5
Irritation test	3.2.2	4.6.1.2	5
Ozone resistance test	3.2.2	4.6.1.3	5
Corrosion	3.5.3	4.6.4	1
Impact test	3.5.5, 3.6.2.1.3	4.6.6	1
Water intrusion test	3.6.2.3	4.6.12.1	4
Shock test	3.5.4	4.6.5	1
Self-extinguishing test	3.2.4	4.6.8	5
Seam strength test	3.6.2.2	4.6.15	6

4.3.2 Sampling for first article inspection. Sampling shall be as specified in 4.3.2.1 through 4.3.2.4.

4.3.2.1 Sampling for rubber compound tests. For each first article inspection of apparatuses, samples of facepieces, breathing tubes, and head harnesses shall be furnished for testing as specified in 4.6.1.1. In order to avoid unnecessary destruction of parts, the samples shall be pieces of rubber compound identical in chemical composition, equivalent in cure, and prepared using identical methods as material used in production. For the physical characteristics tests specified in 4.6.1.1, five sample pieces shall be provided of each part's material and shall be 15.24 by 15.24 by 0.203 ± 0.03 cm (6 by 6 by 0.080 ± 0.005 inch). Dumbbell specimens in accordance with FED-STD-601 shall be cut from the sample pieces. For the irritation test in 4.6.1.2, five samples shall be provided of each part's material. These samples shall be 1.27 by 1.27 cm (0.5 by 0.5 inch) with one reasonably flat surface and shall have no sharp edges or corners. For the ozone resistance test in 4.6.1.3, five samples shall be provided of each part's material. These samples shall be prepared in accordance with ASTM D 1149, Specimen Type A.

4.3.2.2 Sampling for neoprene rubber test. For each first article inspection of apparatuses, five samples of neoprene rubber shall be furnished for testing as specified in 4.6.1.1. The samples shall be neoprene identical in chemical composition, equivalent in cure, and prepared using identical methods as material used in production. The sample pieces shall be 15.24 by 15.24 by 0.203 ± 0.013 cm (6 by 6 by 0.080 ± 0.005 inches). Dumbbell specimens in accordance with FED-STD-601 shall be cut from the sample pieces.

4.3.2.3 Sampling for coated fiberglass cloth test. For each first article inspection apparatuses, five samples of coated fiberglass cloth shall be furnished for testing as specfied in 4.6.8. In order to avoid unnecessary destruction of

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parts, the samples shall be pieces of coated fiberglass cloth identical in chemical and physical composition and prepared using identical methods as material used in production.

4.3.2.4 Sampling for seam strength test. For each first article inspection of apparatuses, six samples shall be furnished for testing as specified in 4.6.15.2.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in 4.4.2.1 through 4.4.2.6 (see 6.3).

4.4.1 Lot. For the purpose of quality conformance inspection and test sampling, a lot is defined as all the apparatuses produced in one facility, using the same production processes and materials, and being offered for delivery at one time. A lot shall consist of not greater than 1300 apparatuses or subcomponents for 1300 apparatuses at any one time.

4.4.2 Sampling for quality conformance inspection. Sampling for quality conformance inspection shall be as specified in 4.4.2.1 through 4.4.2.6.

4.4.2.1 Sampling for visual examination. As a minimum, the contractor shall randomly select a sample quantity from each lot of completed apparatuses in accordance with table VI and examine them in accordance with table VII for the defects listed in table VIII. If one or more defects are found in any sample, the entire lot represented by the sample shall be rejected. The contractor has the option of screening 100 percent of the lot for the defective characteristic(s) or providing a new lot which shall be examined in accordance with the sampling plan contained herein. The contractor shall maintain for a period of three years after contract completion all records of inspections, tests, and any resulting rejections.

TABLE VI. Sampling for visual examination.

Lot size	Critical defects	Major defects	Minor defects
2 to 8	All	All	5
9 to 50	13	8	5
51 to 90	13	8	7
91 to 150	13	12	11
151 to 280	20	19	13
281 to 500	29	21	16
501 to 1200	34	27	19
1201 to 1300	42	35	23

TABLE VII. Visual examinations.

Examination	Requirement	Test method
Assembly	3.3	4.5.2
Construction	3.4	4.5.1
Color	3.8	4.5.1
Finish	3.9	4.5.1
Workmanship	3.11	4.5.1

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TABLE VIII. Classification of defects.

Categories	Defects
Critical: 1 2 3	Breathing apparatus incomplete, component parts missing or improperly assembled as specified. Combination valve discs or seats defective or damaged. Facepiece buckles not operable or as specified.
Major: 101	Rubber materials not free from pits, bumps, nonfills, porous areas, blisters, stickey areas, air traps, flow cracks, cut holes, and tears.
Minor: 201	Breathing apparatus not clean, free of dirt, dust, oil, grease, or extraneous foreign materials.

4.4.2.2 Sampling for group A tests. As a minimum, the contractor shall randomly select a sample quantity from each lot of completed apparatuses in accordance with table IX sampling plan A and examine them in accordance with table X. If one or more defects are found in any sample, the entire lot represented by the sample shall be rejected. The contractor has the option of screening 100 percent of the lot for the defective characteristic(s) or providing a new lot which shall be examined in accordance with the sampling plan contained herein. The contractor shall maintain for a period of three years after contract completion all records of inspections, tests, and any resulting rejections.

TABLE IX. Sampling for group A, B, C, and D tests.

	Sample size			
Lot size	Plan A	Plan B	Plan C	Plan D
2 to 8	3	5	All	All
9 to 25	3	5	8	13
26 to 50	5	5	8	13
51 to 90	6	7	8	13
91 to 150	7	11	12	13
151 to 280	10	13	19	20
281 to 500	11	16	21	29
501 to 1200	15	19	27	34
1201 to 1300	18	23	35	42

TABLE X. Group A test.

Test	Requirement	Test method
Weight	3.7	4.6.13
Threads	3.2.5	4.5.2
Components	3.6	4.5.1, 4.5.2
Identification	3.10	4.5.1
Lens examination	3.6.1.2	4.5.3

4.4.2.3 Sampling for group B tests. As a minimum, the contractor shall randomly select a sample quantity from each lot of completed apparatuses in accordance with table IX sampling plan B and test them in accordance with table XI. If one or more defects are found in any sample, the entire lot represented by the sample shall be rejected. The contractor has the option of screening 100 percent of

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the lot for the defective characteristic(s) or providing a new lot which shall be tested in accordance with the sampling plan contained herein. The contractor shall maintain for a period of three years after contract completion all records of inspections, tests, and any resulting rejections.

TABLE XI. Group B tests.

Test	Requirement	Test method
Lens tests		
Visual characteristics test	3.6.1.2.1	4.6.2.1
Light transmission test	3.6.1.2.1	4.6.2.4
Resistance to gas flow	3.5.1	4.6.3
Leakage tests		
Apparatus body test	3.6.2	4.6.9.1
Facepiece test	3.6.1	4.6.9.2
Relief valve test	3.6.2.2.1	4.6.9.3
Breathing bag test	3.6.2.2	4.6.9.4
Timer loudness and duration test	3.6.2.3	4.6.12.2
Breathing tube test	3.6.1.9	4.6.11
Breathing bag volume test	3.6.2.2	4.6.14
Timer accuracy test	3.6.2.3	4.6.12.3

4.4.2.4 Sampling for group C tests. As a minimum, the contractor shall randomly select a sample quantity from each lot of completed apparatuses in accordance with table IX sampling plan C and test them in accordance with table XII. If one or more defects are found in any sample, the entire lot represented by the sample shall be rejected. The contractor has the option of screening 100 percent of the lot for the defective characteristic(s) or providing a new lot which shall be tested in accordance with the sampling plan contained herein. The contractor shall maintain for a period of three years after contract completion all records of inspections, tests, and any resulting rejections.

TABLE XII. Group C tests.

Test	Requirement	Test method
Lens test Cylinder power test	3.6.1.2.2	4.6.2.2

4.4.2.5 Sampling for group D tests. As a minimum, the contractor shall randomly select a sample quantity from each lot of completed apparatuses in accordance with table IX sampling plan D and test them in accordance with table XIII. If one or more defects are found in any sample, the entire lot represented by the sample shall be rejected. The contractor has the option of screening 100 percent of the lot for the defective characteristic(s) or providing a new lot which shall be tested in accordance with the sampling plan contained herein. The contractor shall maintain for a period of three years after contract completion all records of inspections, tests, and any resulting rejections.

TABLE XIII. Group D tests.

Test	Requirement	Test method
Lens test Strength and impact resistance test	3.6.1.2	4.6.2.3

4.4.2.6 Performance. Two apparatus shall be pulled from each 650 units offered for delivery, regardless of lot size, and tested as specified in 4.6.10.

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4.5 Examinations.

4.5.1 Visual. Sample apparatuses shall be visually examined for conformance to the requirements of 3.4, 3.6.1.1.1, 3.6.1.3, 3.6.1.6, 3.6.1.6.1, 3.6.1.8.1, 3.6.2.1.2, 3.8, 3.9, 3.10, and 3.11.

4.5.2 Dimensions. Sample apparatuses shall be examined for conformance to the requirements of 3.2.5, 3.3, 3.6.1, 3.6.2, 3.6.2.1.1, 3.6.2.1.3, and 3.6.3.

4.5.3 Lens examination. The lens shall be examined for surface imperfections and other markings as specified in 3.6.1.2.1. The lens shall be examined for scratches and striae by being placed in the axis of a magnifying lens system and illuminated from the back by an incandescent lamp placed to fully illuminate the lens. Striae will appear as dark lines; scratches will appear as illuminated lines.

4.6 Test methods. Tests shall be as specified in 4.6.1.1 through 4.6.15.

4.6.1 Rubber compound tests. Rubber compound tests shall be as specified in 4.6.1.1 through 4.6.1.3.

4.6.1.1 Physical characteristics. Rubber compound and neoprene rubber samples shall be selected as specified in 4.4.2.1 and 4.4.2.2, respectively, and shall be tested in accordance with ASTM D 412. Oven aging shall consist of exposing the samples to an air oven at 100 ± 1°C (212 ± 2°F) for 24 hours or at 70 ± 1°C (158 ± 2°F) for 168 hours. Samples of nitrile rubber shall conform to the requirements of 3.2.2 and samples of neoprene rubber shall conform to the requirements of 3.2.3.

4.6.1.2 Rubber irritation tests. Rubber compound samples shall be selected as specified in 4.4.2.1 and shall be tested for conformance to 3.2.2 by the following procedure.

4.6.1.2.1 Supervision. The rubber irritation tests shall be performed only under the supervision of a qualified physician in the presence of a designated Government representative.

4.6.1.2.2 Procedure. Five test subjects shall have a sample from each part (see 4.4.2.1) firmly affixed with surgical adhesive tape to the inner side of their upper arm. The samples shall be worn for 24 hours and then shall be removed. Upon removal of the samples, 24 hours after removal, and 48 hours after removal, the skin in the sample contact areas shall be examined for reactions as specified in table XIV. If a plus 1 or higher reaction occurs during the 24-hour period when the samples are being worn, the test shall be immediately discontinued.

4.6.1.2.3 Test results. Any reaction of plus 1 or higher shall constitute failure of the rubber compound in the part or parts causing the reaction.

TABLE XIV. Severity of skin reactions.

Rating	Reaction
0	No reaction
+1	Redness
+2	Redness and swelling
+3	Pimples and blisters
+4	Ulceration or tissue destruction

4.6.1.3 Rubber ozone resistance. Rubber compound samples shall be selected as specified in 4.3.2.1 and shall be tested for conformance to 3.2.2 in accordance with ASTM D 1149. Samples shall be tested at 2.5 percent elongation for 70 hours at an ozone partial pressure of 50 mPa. Samples shall be observed at a magnification of 7x at the end of the test period and shall exhibit no visible cracking.

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4.6.2 Lens tests. Lens tests shall be as specified in 4.6.2.1 through 4.6.2.4.

4.6.2.1 Visual characteristics test. Sample lenses shall be tested for resolving power by means of a telescope with a magnifying power of at least six, resolving at least 15 arc seconds, stopped down to an apperture of 10 mm (0.4 inch) in accordance with NIST NBS Circular Special 374, test pattern 14, using a distance of 4.572 meters (15 feet). The lens shall conform to the requirements of 3.6.1.2.1.

4.6.2.2 Cylinder power test. Sample lenses shall be tested for cylinder power through the use of a lensometer or other instrument. The lens axis shall be located and marked. The cylinder power shall be determined at sufficient locations on the surface to indicate the positions of the maximum values. The lens shall conform to the requirements of 3.6.1.2.2.

4.6.2.3 Strength and impact resistance test. Sample lenses shall be mounted normally in facepieces including the retaining rings. The facepiece shall be mounted on a wooden block so that the outer surface of the lens faces upward. The block shall be shaped to support the lens seal and the retaining rings, but the block shall not support the lens. The whole assembly shall be on a flat, horizontal surface. A 25.4-mm (1-inch) diameter steel ball weighing approximately 68 grams (2.4 ounces) shall be freely dropped from a height of 1.27 meters (50 inches) onto the center of the lens. The sample lenses shall conform to 3.6.1.2.

4.6.2.4 Light transmission test. The luminous transmittance of sample lenses shall be determined in accordance with ASTM D 1003. The transmission shall be as specified in 3.6.1.2.1.

4.6.3 Resistance to breathable gas flow. Sample apparatuses shall be tested for resistance to breathable gas flow which shall not exceed the requirement in 3.5.1. For the purpose of this test, a dummy canister with the known resistance of 18 mm (0.71 inch) water column height at a breathable gas flow rate of 85 liters per minute (L/min) (3 cubic feet per minute (ft^3/min)) shall be inserted into the apparatus. The supply of breathable gas shall be provided at the exhalation port in the facepiece and shall be at a flow rate of 85 L/min (3 ft^3/min).

4.6.4 Corrosion. Each sample apparatus shall be placed upright in a salt spray cabinet and shall be subjected to a salt spray atmosphere for 100 hours with a 5 percent salt solution in accordance with FED-STD-151. Sample apparatuses shall be as specified in 3.5.3.

4.6.5 High impact shock. Sample apparatuses shall be subjected to shock testing for grade A, class 1, type A lightweight equipment in accordance with MIL-S-901. The mounting fixture shall be an enclosed aluminum alloy box with internal dimensions of 72.1 by 21.6 by 59.5 cm (28.375 by 8.5 by 23.4375 inches (in)). The aluminum alloy sheet shall be 0.20 cm (0.08 in) thick. The apparatus body shall be placed in this container in accordance with Drawing 804-5773272 plan view A-ASF; the container shall remain in a horizontal position during testing. Sample apparatuses shall be as specified in 3.5.4.

4.6.6 Impact. Sample apparatus bodies in both the standby and actuated positions shall be dropped onto a concrete surface. In the standby position a dummy canister shall be inserted into the body, and in the actuated position a dummy canister that shall be of the same weight as an actual canister in accordance with MIL-C-17671 shall be inserted into the body. The lowest point of the body shall be 0.91 meter (3 feet) above the concrete in each orientation. The orientations from which each body in both positions shall be dropped shall be as follows:

(a) Vertical with the retaining assembly at the lowest point.
(b) Horizontal with the convex side of the canister guard at the lowest point.

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(c) Horizontal with the convex side of the canister guard at the highest point.

Sample apparatus bodies shall be as specified in 3.5.5 and 3.6.2.1.3.

4.6.7 Wear. Sample apparatus bodies shall have a spent canister loaded and unloaded for 100 cycles. A cycle shall consist of inserting the dummy canister in the standby position, moving bail to the actuated position, returning to the standby position, and removing the canister utilizing the tab in the retaining assembly. Sample bodies shall be as specified in 3.5.6.

4.6.8 Self-extinguishing. The coated fiberglass cloth samples as specified in 4.4.2.3 shall be prepared and tested for flameout in accordance with FED-STD-191, test 5903. The cloth samples shall be as specified in 3.2.4.

4.6.9 Leakage tests. Leakage tests shall be conducted as specified in 4.6.9.1, 4.6.9.2, and 4.6.9.3 to determine conformance to the requirements of 3.5.2.

4.6.9.1 Apparatus body procedure. Each sample apparatus body shall have a dummy canister inserted to form a gas-tight seal with the plunger assembly. The inhalation and exhalation tube fittings shall be sealed gas-tight. The relief valve shall be prevented from venting. The apparatus body shall be subjected to a pressure of 25.4 cm (10 inches) of water column height (manometer diameter shall be a minimum of 3.175 mm (0.125 inch)) to inflate the breathing bags. One minute after reducing the pressure to 12.7 cm (5 inches) of water column height, the pressure drop shall be checked for conformance to 3.6.2. The relief valve shall then be allowed to vent. The apparatus body shall be inflated with sufficient pressure to vent the relief valve. Air flow shall be stopped allowing breathing apparatus to deflate freely. After pressure has stabilized for 30 seconds, the pressure drop shall be checked for conformance to 3.6.2 relief valve setting pressure requirements.

4.6.9.2 Facepiece assembly procedure. Each sample facepiece assembly shall have the inhalation and exhalation tube fittings sealed and shall be placed over a headform to form gas-tight seal. A vacuum of 15.2 cm (6.0 inches) of water column height shall be applied to the inside of the facepiece. An in-leakage test material shall be applied to the outside of the entire facepiece assembly. The percent in-leakage shall be checked after 15 seconds into the test for conformance to 3.6.1.

4.6.9.3 Relief valve procedure. Prior to assembly of relief valve into breathing bags, insert valve into a test fixture and apply 10.16 cm (4 inches) water negative pressure to the inner side of the valve. Leakage of the valve shall conform to 3.6.2.2.1.

4.6.9.4 Bag volume. The breathing bags shall be drawn flat by applying a vacuum of 4 to 6 inches of water. The bags shall then be inflated and the volume measured. The volume in the bag shall be 5 liters (0.18 ft^3) minimum when the relief valve opens (see 3.6.2.2).

4.6.10 Performance. Following the tests specified in 4.6.3 and 4.6.9, each sample apparatus shall be subjected to a performance test. The performance test shall be as specified in 4.6.10.1. Test samples shall be taken at the point where the breathable gas enters the facepiece after exiting the apparatus body. The sample apparatus shall conform to 3.5.7. The duration of operation shall be determined by the carbon dioxide levels exceeding the requirement of 3.5.7 or the breathing resistance exceeding 95 mm (3.74 inches) of water column height.

4.6.10.1 Automatic breathing metabolic simulator method. For this test, a canister shall be properly inserted into the sample apparatus. The facepiece shall be connected to the headform of an automatic breathing metabolic simulator. After actuating the canister, the apparatus shall be tested for at least 50 minutes using the following metabolic parameters:

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(a)	Oxygen consumption rate	- 1.35 L/min (0.048 ft^3/min)
(b)	Carbon dioxide production rate	- 1.15 L/min (0.041 ft^3/min)
(c)	Ventilation rate	- 32.00 Ll/min (1.13 ft^3/min)
(d)	Respiratory frequency	- 22.0 breaths per minute
(e)	Exhaled humidity	- 80 percent saturated
(f)	Exhalation temperature	- 35°C (95°F)

4.6.11 Breathing tube. Each sample breathing tube with fitting shall be immersed in water and subjected to an internal air pressure of 43.5 kilopascals (kPa) (5 pounds per square inch gauge). No leakage shall occur while pressure is applied for a minimum of 10 seconds. The breathing tube shall conform to the requirements of 3.6.1.9.

4.6.12 Timer test procedures. Timer test procedures shall be as specified in 4.6.12.1 through 4.6.12.4.

4.6.12.1 Water intrusion. Each sample timer assembly shall be placed with knob up under a 1.59 cm (0.625 inch) inside diameter pipe and shall be subjected to water flow from that pipe for 60 minutes at a rate of 2.88 L/min (0.75 gallons per minute (gal/min)). At the conclusion of the hour, the sample timer assemblies shall be inspected for conformance to 3.6.2.3.

4.6.12.2 Loudness. Each sample timer assembly shall be placed so a sound level meter is 30.5 cm (12 inches) directly above the bottom of the timer's knob. When the alarm bell is actuated, both the loudness and duration shall be measured. The bell shall perform as specified in 3.6.2.3.

4.6.12.3 Accuracy. Each sample timer shall be wound, set at 1 hour, and activated. The timer shall perform as specified in 3.6.2.3.

4.6.12.4 Knob strength. Each sample timer knob shall be placed on a shaft of equal dimensions as the actual shaft on the timer assembly. The material composition of the test shaft may be different from that of the actual timer assembly shaft. A shaft torque of 35 inch-pounds shall be applied to the timer knob. After testing, the knob shall be inspected for conformance to 3.6.2.3.

4.6.13 Weight. Each complete sample apparatus shall be tested for conformance to 3.7.

4.6.14 Breathing bag volume. Each sample apparatus body shall have the plunger assembly and inhalation and exhalation tube fittings sealed gas-tight. The breathing bags shall be drawn flat by a vacuum of 12.7 cm (5 inches) water column height. The bags shall be inflated. The volume of gas required to just open the relief valve shall be measured and shall conform to 3.6.2.2.

4.6.15 Breathing bag seam strength tests. From the six breathing bag samples, four shall be tested for cyclic fatigue as specified in 4.6.15.1 and two shall be tested for seam tensile peel strength as specified in 4.6.15.2.

4.6.15.1 Cyclic fatigue tests. Each sample breathing bag shall be assembled on the apparatus with the facepiece removed and a dummy canister installed. With the addition of air to the plunger valve, the bag shall be inflated to achieve opening pressure of the relief valve, followed by the withdrawal of air from the elbow fitting to reduce the pressure in the bag to not greater than 1.0 inch of water. This cycle shall be repeated 250,000 times without failure of the bag seam.

4.6.15.2 Seam tensile peel tests. Representative 1.0-inch wide samples shall be cut from inboard and outboard seams of the left and right sections of the two sample bags (total of four samples each bag). Each sample shall be preconditioned at 121 ± 2°C (250 ± 3.6°F) for 5 minutes, then tested for tensile peel strength in accordance with ASTM D 413 while maintaining the sample exposure temperature at 121

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± 2°C (250 ± 3.6°F). (Note: If sewn reinforcement construction is used, samples may be sewn after die cutting to avoid unraveling the seam.) The tensile strength shall be as specified in 3.6.2.2.

4.7 Inspection of packaging. Sample packs, and the inspection of packaging (preservation, packing and marking) for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging or preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 General.

5.1.1 Navy fire-retardant requirements.

5.1.1.1 Lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Level A and B	- Type II - weather resistant. Category 1 - general use.
Level C	- Type I - nonweather resistant. Category 1 - general use.

5.1.1.2 Fiberboard. Fiberboard used in the construction of interior (unit and intermediate) and exterior fiberboard boxes including interior packaging forms shall conform to the class-domestic/fire retardant or class-weather resistant/fire retardant materials requirements as specified (see 6.2), of ASTM D 4727.

5.2 Preservation. Preservation shall be level A, C or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. The breathing apparatus (see 3.4) shall be cleaned and dried by a process and procedure that will ensure that the apparatus is free of all petroleum products, hydrocarbons, and other foreign material that may affect the operation of the apparatus. Cleaning and drying shall be in accordance with MIL-P-116. Unless otherwise specified (see 6.2), the selection of the cleaning process and the drying procedure shall be at the contractor's option.

5.2.1.2 Unit protection. Each assembled (ready-for-use) breathing apparatus shall be individually unit protected within a transparent bag in accordance with MIL-P-116, method IC-1. Interior wrapping or cushioning material shall be provided, as necessary, to prevent striae and scratches to the facepiece lens, and puncture or damage to the protective greaseproof, waterproof bag. Each IC-1 unit bag shall be supplemented with a snug-fitting, class-weather-resistant, fire retardant fiberboard box conforming to PPP-B-636. Unless otherwise specified (see 6.2), box options such as type, variety, grade and style shall be the choice of the contractor. Cushioning or blocking shall be provided at the contractor's option. Use of loose-fill material is prohibited as cushioning, filler, stuffing or dunnage material. Cushioning or wrapping materials as applicable shall have properties (characteristics) for resistance to fire. Blocking shall be by the use of fiberboard forms and be of the same material as the unit box. Box closure and reinforcing shall be in accordance with the appendix to the box specification, method V. Unless otherwise specified (see 6.2), overpacking is not required for shipment.

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5.2.2 Level C. Preservation shall be as specified for level A except that (a) The unit protection shall be method IC-3, bag closure by twist tie, or pressure-fit, and the leak and heat-seal tests are not required and (b) the unit fiberboard box shall be class-domestic/fire retardant (see 5.1.1.2) or when specified (see 6.2) class-domestic. Box closure shall be in accordance with the appendix to the box specification, method I, using pressure sensitive tape. Unless otherwise specified (see 6.2), overpacking in exterior shipping containers is not required for levels B, C, and commercial packing requirements.

5.2.3 Commercial. Commercial packaging (cleaning, preservation, cushioning, and unit pack) shall be in accordance with ASTM D 3951.

5.3 Packing. Packing when required (see 6.2), shall be level A, B, C, or commercial as specified (see 6.2).

5.3.1 General requirements for levels A, B, and C. Containers selected (see 5.3.2) shall be of minimum weight and cube consistent with the protection required, of uniform size, and contain identical quantities.

5.3.2 Levels A, B, and C containers. Breathing apparatuses preserved as specified (see 5.2) shall be packed when specified (see 5.3) in exterior shipping containers in accordance with table VII of MIL-STD-2073-1 for the level of packing specified (see 5.3) and herein. Unless otherwise specified (see 6.2), container selection including container options shall be the contractor's option.

5.3.2.1 Caseliners, closure and gross weight.

5.3.2.1.1 Caseliners. Unless otherwise specified (see 6.2), level A shipping containers containing apparatuses preserved level C or commercial shall be provided waterproof caseliners in accordance with MIL-STD-2073-1.

5.3.2.1.2 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that weather-resistant fiberboard boxes shall be closed in accordance with method V and reinforced with nonmetallic or tape banding and domestic or fire retardant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

5.3.2.1.3 Skidded containers. Wood, plywood, and cleated type containers exceeding 90.72 kg (200 pounds) gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1, appendix F or the applicable container specification and appendix thereto as applicable.

5.3.3 Commercial. Apparatuses preserved as specified (see 5.2) shall be packed for shipment in accordance with ASTM D 3951 and herein.

5.3.3.1 Container modification. Shipping containers exceeding 90.72 kg (200 pounds) gross weight shall be provided with a minimum of two 7.62 by 10.16 cm (3- by 4-inch) nominal wood skids laid flat, or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment and storage.

5.4 Palletized unit loads. When specified (see 6.2), containers shall be palletized in accordance with MIL-STD-2073-1.

5.5 Marking.

5.5.1 Levels A, B, C, and commercial. In addition to any special marking required (see 6.2), and 3.10 interior (unit) packs, shipping containers and palletized unit loads shall be marked for shipment, stowage, and storage in accordance with MIL-STD-2073-1, appendix F.

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6. NOTES

6.1 Intended use. The oxygen breathing apparatus is intended to be used with MIL-C-17671 quick-starting canister to provide fire fighting, damage control and other personnel with safe and sufficient breathing air for entry into spaces where the ambient air is contaminated with toxic gases or deficient in oxygen.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DODISS to be cited in the solicitation, and if required the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) When first article is required (see 3.1).
(d) When fire-retardant requirements are not required (see 5.1.1).
(e) Levels of preservation and packing required (see 5.2 and 5.3).
(f) Cleaning and drying selection if other than contractor's choice (see 5.2.1.1).
(g) Box options if other than contractor's choice (see 5.2.1.2 and 5.3.2).
(h) When overpacking is required (see 5.2.1.2, 5.2.2 and 5.3).
(i) When class-domestic box is required (see 5.2.2).
(j) When caseliners are not required (see 5.3.2.1.1).
(k) Palletization when required (see 5.4).
(l) Special marking required (see 5.5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID number	DID title	Suggested tailoring
4.4	DI-NDTI-80809	Test, Inspection reports	-------

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a reproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.5 Sub-contracted material and parts. The packaging and preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Subject term (key word) listing.

Firefighting equipment
Gas mask
Life support devices
Neoprene-coated fabric
Rescue equipment
Rubber compound
Self-extinguishing fabric
Skin tests, rubber irritation
Tubes

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - EA
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

Preparing activity:
Navy - SH
(Project 4240-0574)

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